Equivalence Partitioning for Deposit Amounts

PROBLEM STATEMENT

A deposit system accepts amounts from INR 1.00 to INR 500,000.00, including both limits. The amount may contain zero, one, or two decimal places. Write a program that checks whether the input is a valid amount. Reject negative amounts, zero, values above the limit, and amounts with more than two decimal places.

Input Format / Test Setup

One amount as text.

Expected Output / Deliverable

VALID or INVALID.

Sample Case or Evaluation Requirement Input:

2500.50;

expected output: VALID.

Testing and Evaluation Requirements Identify valid and invalid equivalence classes, including zero, negative, too many decimal places, and values above the maximum.

ALGORITHM

1. Start the program.
2. Store the eight deposit amounts for testing.
3. Read each amount one by one.
4. Check whether the amount contains only valid numeric characters.
5. Check whether the amount has more than two decimal places.
6. If the amount is invalid or has more than two decimal places, display INVALID.
7. Otherwise, check whether the amount is between ₹1 and ₹5,00,000, including both limits.
8. If the amount is within the range, display VALID; otherwise, display INVALID.
9. Repeat the steps until all eight test cases are checked.
10. Stop the program.

PROGRAM

```
    IDENTIFICATION DIVISION.
    PROGRAM-ID. DEPOSIT-TEST.

    DATA DIVISION.
    WORKING-STORAGE SECTION.
    01 TEST-DATA.
      05 AMOUNT-TEXT PIC X(10) OCCURS 8 TIMES.
    01 I PIC 99.
    01 AMOUNT-VALUE PIC 9(9)V99.
    01 DECIMAL-COUNT PIC 99.
    01 DOT-FOUND PIC X.
    01 VALID-FLAG PIC X.
    01 NUM-LEN PIC 99.
    01 CHAR-POS PIC 99.

    PROCEDURE DIVISION.
      MOVE "2500.50" TO AMOUNT-TEXT(1)
      MOVE "1"      TO AMOUNT-TEXT(2)
      MOVE "500000.00" TO AMOUNT-TEXT(3)
      MOVE "0"      TO AMOUNT-TEXT(4)
      MOVE "-100"    TO AMOUNT-TEXT(5)
      MOVE "500001"  TO AMOUNT-TEXT(6)
      MOVE "10.123"  TO AMOUNT-TEXT(7)
      MOVE "abc"    TO AMOUNT-TEXT(8)
```

```
PERFORM VARYING I FROM 1 BY 1
  UNTIL I > 8

  MOVE 'Y' TO VALID-FLAG
  MOVE 'N' TO DOT-FOUND
  MOVE 0 TO DECIMAL-COUNT

  COMPUTE NUM-LEN =
    FUNCTION LENGTH(
    FUNCTION TRIM(AMOUNT-TEXT(I)))

  PERFORM VARYING CHAR-POS FROM 1 BY 1
    UNTIL CHAR-POS > NUM-LEN
    IF AMOUNT-TEXT(I)(CHAR-POS:1) = '.'
      MOVE 'Y' TO DOT-FOUND
    ELSE
      IF DOT-FOUND = 'Y'
        ADD 1 TO DECIMAL-COUNT
      END-IF
    END-IF
  END-PERFORM

  IF DECIMAL-COUNT > 2
    MOVE 'N' TO VALID-FLAG
  END-IF
```

```
      IF FUNCTION TEST-NUMVAL(
        FUNCTION TRIM(AMOUNT-TEXT(I))) NOT = 0
        MOVE 'N' TO VALID-FLAG
      ELSE
        IF FUNCTION NUMVAL(
           AMOUNT-TEXT(I)) < 1 OR
          FUNCTION NUMVAL(
           AMOUNT-TEXT(I)) > 500000
           MOVE 'N' TO VALID-FLAG
        END-IF
      END-IF

      IF VALID-FLAG = 'Y'
        DISPLAY FUNCTION TRIM(AMOUNT-TEXT(I))
           " : VALID"
      ELSE
        DISPLAY FUNCTION TRIM(AMOUNT-TEXT(I))
           " : INVALID"
      END-IF

    END-PERFORM

    STOP RUN.
```

OUTPUT

```
Output:
2500.50 : VALID
1 : VALID
500000.00 : VALID
0 : INVALID
-100 : INVALID
500001 : INVALID
10.123 : INVALID
abc : INVALID
```

RESULT

The COBOL program was executed successfully. It checked all eight deposit amounts and displayed VALID for correct amounts and INVALID for incorrect amounts.